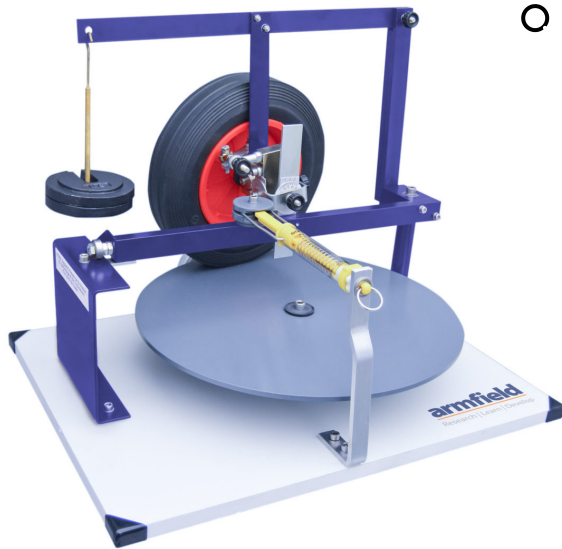




SD-1.57 – Castor, Camber and King Pin Inclination

Bench top unit simulating vehicle front steering arrangement

PRODUCT CODE: SD-1.57

Categories: Educational, SD-1.57-Castor Camber And King Pin Inclination, SV - Theory Of Machines, SV Series - Statics & Vibrations

Tags: Automotive , Camber And King Pin Inclination , Castor , Castor Angle , Cornering Forces , Geometry , King Pin Inclination , SD-1.57 , Steering , Toe In , Toe Out , Vernier Scale

DESCRIPTION

Students often experience difficulty in understanding the principles of front wheel set-up on road vehicles, since the geometry involves angles in three dimensions.

This apparatus shows precisely how castor, camber and king pin inclination are set up and how centre point steering is achieved.

A wheel and means of altering the offset are included so that everything affecting centre point steering can be varied. Vernier scales measure the castor angles and king pin inclination, while toe in and camber angles are measured from a longitudinal and vertical datum in the way in which they would be determined on a real vehicle.

A hand driven rotating turntable simulates a rolling road, and may be turned at a slow enough speed to study wheel stability under running conditions.

TECHNICAL SPECIFICATIONS

- Ø250mm Running wheel
- Ø380mm Turntable
- $\pm 5^\circ$ Vernier scale
- $\pm 40^\circ$ Protractor scale

FEATURES & BENEFITS

- Bench top unit simulating vehicle front steering arrangement
- Very visual unit
- Rolling road can be turned manually
- Variation of Castor, Camber, King Pin Inclination

- Accurate adjustment using protractors and flexible joints
- Many adjustments can be made to vary results

Experimental Content

- To show how castor, camber and king pin inclination are set up
- To show how the steering linkage is related to toe-in
- To demonstrate the effect of wheel offset
- To observe the dynamic stability of the wheel under running conditions
- To measure the forces on the steering link
- To show how the weight of the car affects self centring

Ordering Specification

- 1 x **SD-1.57** assembly
- 1 x Combination spanner
- Hex wrench
- Packing list
- Test sheet

Series includes:

Theory of Machines

- SD-1.50 Wheel and Axle
- SD-1.51 Wheel and Differential Axle Apparatus
- SD-1.52 Efficiency of Screw Threads
- SD-1.53 Gear Tooth Form Apparatus
- SD-1.54 Cam and Follower Apparatus
- SD-1.55 Simple Flywheel
- SD-1.56 Epicyclic Gear Train
- SD-1.57 Castor, Camber, and King Pin Inclination
- SD-1.58 Relation Between Angular and Linear Speeds
- SD-1.59 Disc Brake Apparatus
- SD-1.60 Static & Dynamic Balancing Apparatus
- SD-1.61 Governor Apparatus
- SD-1.62 Gear Assembly Unit Combined Drives
- SD-1.63 Single Epicyclic Gear Train

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 0.24m³

Gross Weight: 33kg



Dimensions

Length: 0.500m

Width: 0.400m

Height: 0.370m

Order Codes

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